



EXAMPLE GAUSSIAN PATHS

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$$I(\beta) = \frac{\dot{\mu}_{\beta}^2 + 2\dot{\sigma}_{\beta}^2}{\sigma_{\beta}^2}$$

► **Example:** Suppose $\pi_{\beta} = \mathcal{N}(\mu_{\beta}, \sigma_{\beta}^2)$ then the Fisher information equals



If $\sigma_\beta \equiv \sigma$ is constant then:

$$I(\beta) = \frac{\dot{\mu}_{\beta}^2}{\sigma^2}$$

Changes in location correspond to linear changes in mass



If $\mu_\beta = \mu$ is constant then:

$$I(\beta) = \frac{\dot{\sigma}_{\beta}^2}{\sigma_{\beta}^2} = \left(\frac{\mathrm{d}}{\mathrm{d}\beta} \log \sigma_{\beta} \right)^2$$

► Changes in scale correspond to logarithmic changes in mass independent of μ

► The distribution changes more rapidly when σ is small for the unit change in μ

EXAMPLE GAUSSIAN PATHS

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- ▶ Changes in location correspond to linear changes in mass
- ▶ The distribution changes more rapidly when σ is small for the unit change in μ

- ▶ If $\mu_\beta = \mu$ is constant then:

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CALCULUS OF ANNEALING PATHS